Total Printed Pages: 03

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MAB-101
Examination –Dec- 2023
B.Tech. I Sem : Common to all Branches
Linear Algebra and Calculus

Time : 2 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:
Part a & b – Objective type/Fill in the blanks/One sentence answer.
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

			BL	CO
Q.1	(a)	State Maclaurins theorem for expansion of functions.	02	1	1
	(b)	The maximum value of a cosx + b sinx is $(a)\,a+b\;(b)\,a-b\;(c)\,\lvert a\rvert+\lvert b\rvert\;(d)\,\sqrt{a^2+b^2}$	02	3	1
	(c)	Expand $Sin\,x$ in powers of $\left(x-\pi/2\right)$	03	2	1
		OR			
		Find the radius of curvature of $y=4\,Sin\,x-Sin\,2x$ at $x=\pi/2$	03	3	1
	(d)	Find the radius of curvature at the point 't' for the following $x=a\,Cos^3t,\quad y=a\,Sin^3t$	05	3	1
		OR			
		Show that the function $f(x,y)=x^3+y^3-63(x+y)+12xy$ is maximum at $(-7,-7)$ and minimum at $(3,3)$.	05	4	1
Q.2	(a)	State the Euler's Theorem of Homogeneous functions in partial differentiation.	02	1	2
	(b)	Define Absolute, Relative and percentage Error.	02	2	2
	(c)	If $U=Sin^{-1}\dfrac{x^2+y^2}{x+y}$ Show that $x\dfrac{\partial u}{\partial x}+y\dfrac{\partial u}{\partial y}=\tan U$	03	3	2
		OR			
		If $Z=x^2\tan^{-1}y/x-y^2\tan^{-1}x/y$ Prove That $\dfrac{\partial^2 z}{\partial y\,\partial x}=\dfrac{x^2-y^2}{x^2+y^2}$	03	3	2

(d) If $U = u\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$, Show that

$$x^2\frac{\partial u}{\partial x} + y^2\frac{\partial u}{\partial y} + z^2\frac{\partial u}{\partial z} = 0$$

OR

If $u = f(r)$ where $r^2 = x^2 + y^2$

Show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r}f'(r)$

05	3	2

05	3	2

Q.3 (a) The Limit of the series

$$\lim_{n\to\infty}\left[\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \cdots + \frac{1}{n+n}\right] \text{ is}$$

(a) $\log(1+n)$ (b) $\log 2$ (c) $\log 4$ (d) $\log 3$

(b) The Value of $\int_0^1\int_0^x\int_0^{x+y} dx\,dy\,dz$ is

(a) 1 (b) 1/3 (c) 1/4 (d) 1/2

(c) Evaluate $\int_0^1 \frac{\log(1+x)}{1+x^2}\,dx$

OR

Evaluate $\lim_{n\to\infty}\sum_{r=1}^{n-1}\frac{1}{n}\sqrt{\frac{n+r}{n-r}}$

(d) Evaluate $\int_0^1\int_0^{\sqrt{1+x^2}}\frac{dy\,dx}{1+x^2+y^2}$

OR

Find by Double integration the area enclosed by the pair of curves.

$y = 2 - x$ and $y^2 = 2(2-x)$

02	3	3
02	5	3
03	5	3

03	5	3

05	5	3

05	5	3

Q.4 (a) Define the rank and Adjoint of a Matrix.

(b) State Cayley Hamilton Theorem for Square Matrix.

(c) If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$

Find Adj A and its inverse.

02	1	4
02	1	4
03	3	4

OR

Discuss the consistency of the following system of equations. 03 | 3 | 4

$2x+3y+4z=11, \quad x+5y+7z=15, \quad 3x+11y+13z=25$

If found consistent solve it.

(d) Find the Eigen value of the Matrix. 05 | 3 | 4

$$A=\begin{bmatrix}1 & -6 & -4\\ 0 & 6 & 2\\ 0 & -4 & -3\end{bmatrix}$$

Also find only one Eigen Vector.

OR

For what value of λ and μ the equations. 05 | 4 | 4

$2x+3y+5z=9, \quad 7x+3y-2z=8, \quad 2x+3y+\lambda z=\mu$

Have (i) No solution (ii) an unique solution

(iii) an infinite number of solutions.

Q.5 (a) Define the following 02 | 1 | 5

(i) Complete Graph

(ii) Euler Graph

(b) Define Duality in Boolean Algebra $(B, +, *, ')$ 02 | 1 | 5

(c) Prove that if a graph (connected or disconnected) has exactly two vertices of odd degree, there must be a path joining these two vertices. 03 | 2 | 5

OR

State and prove the associative law in Boolean Algebra. 03 | 3 | 5

(d) Solve the following Boolean function 05 | 3 | 5

$X*Y+\overline{X*Z}+X*\overline{Y}*Z\left(X*Y+Z\right)$

OR

Define the following. 05 | 1 | 5

(i) Isomorphism of Graph

(ii) Tree

(iii) Hamiltonian Graph

(iv) Degree of a Vertex

(v) Sub graphs.

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating

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